FS Engine Integration Spec

Connecting the FS Exam Prep mobile app (GUI) to the fs-exam-prep question engine

Version 1.0.0 · 2026-10-21 | Status: approved plan, ready to build | Companion files: Instruction Prompt v1.0.0 (PDF) and bank-seed-20261021.json (data)

1 Vision and ownership

The engine owns the questions. The `fs-exam-prep` repository (Python, v1.0.0) generates practice-question banks with 48 parameterized generators across the 7 official Fundamentals of Surveying (FS) knowledge areas. Banks are JSON files in a **locked schema v1** format: within the v1 line, no field is ever removed or renamed, validation rules are never tightened against previously valid banks, and any breaking change must ship as schema v2 with a migration path.

The app is the wrapper. The FS Prep mobile app (Kotlin / Jetpack Compose, built in Google AI Studio) owns presentation and user state — progress, mastery, achievements, theme. It must stop owning content. Today it hardcodes 32 questions in source code; after this integration it loads engine banks and contains no questions of its own.

Reference implementation. The Flutter fs-exam-app already consumes this exact contract (its docs/CONTRACT.md defines the field mapping and change protocol). Mirror it field-for-field.

2 The payload

One bank file ships with the app in this phase:

Item	Value
File	bank-seed-20261021.json
Generated by	fs-exam-prep engine v1.0.0, CLI: generate --all --n 10 --seed 20261021
Contents	480 questions · 7 knowledge areas · 48 topics · validated by the engine's own validator (schema v1)
Difficulty mix	Level 1: 180 · Level 2: 280 · Level 3: 20
Install path in app	app/src/main/assets/banks/bank-seed-20261021.json (byte-for-byte unchanged)
SHA-256	71ebea4a2c9a15ff31637a063ff03b2af18e1120c01f42faaec7879b014b235a
Refreshing content	Generate a new seed, drop the new file in, nothing else changes.

Bank composition by knowledge area: survey-computations 120 · boundary-law-and-real-property 70 · surveying-processes-and-methods 70 · applied-math-statistics 60 · mapping-processes-and-methods 60 · business-concepts-ethics 50 · surveying-principles-geodesy 50.

3 Contract mapping — engine JSON to app model

Engine JSON field	App Question field	Rule
qid	id	Non-empty; unique per bank (loader enforces)
domain (slug)	domain	Matched against the 7 slugs in §4
difficulty (int 1–3)	difficulty	1 → EASIER · 2 → ABOUT_THE_SAME · 3 → HARDER
stem	questionText	Non-empty
choices	options	Exactly 4 unique non-empty strings
answer_index (0–3)	correctIndex	Grading stays: selectedIndex == correctIndex
explanation	explanation	Non-empty
topic	topicTag	Slug string
—	formulaHighlight	Always null (engine has no such field)
—	questionNumberLabel	Computed at display time, never stored
parameters, seed	(ignored)	Readers may ignore; never shown in UI

4 Taxonomy — the app's 8 domains are replaced by the engine's 7

Slug (id)	Title	Code
applied-math-statistics	Applied Math & Statistics	MATH
boundary-law-and-real-property	Boundary Law & Real Property	LAW
business-concepts-ethics	Business Concepts & Ethics	ETHICS
mapping-processes-and-methods	Mapping Processes & Methods	MAP
survey-computations	Survey Computations	COMP
surveying-principles-geodesy	Surveying Principles & Geodesy	GEOD
surveying-processes-and-methods	Surveying Processes & Methods	PROC

Where the old domains land: PLSS becomes a topic inside boundary-law; Curves becomes topics inside survey-computations (horizontal-curve, vertical-curve); Methods & Equipment maps to surveying-processes-and-methods; Photogrammetry/GIS maps to mapping-processes-and-methods. The old per-domain 'target exam questions' ranges are deleted — they summed past the real exam's 110 questions and matched no official distribution.

5 Loader behavior — non-negotiables

- Load once, lazily, from app assets. Parse with **org.json only** (built into Android). **No new dependencies of any kind.**
- **Validate on load:** bank schema_version == 1; question_count equals the actual number of questions; every question passes the §3 rules (unique non-empty qid, exactly 4 unique non-empty choices, answer_index in 0–3, non-empty stem and explanation).

- **Fail visibly, never silently.** On any violation, show an error state naming the specific problem. There must be no fallback or sample question set anywhere in the app — a wrong answer key is worse than no quiz.
- The QuestionBank object keeps its existing public API — `questions`, `getQuestionsForQuiz`, `getAdaptiveQuestions`, `getMockExamQuestions` — same signatures and selection semantics, now sourced from the loaded bank. Screens and the ViewModel must not need changes.
- **The mock exam becomes honest:** default draw is **110 questions**, distributed proportionally across the 7 knowledge areas, with a time limit of **320 minutes** (the real exam's 5 hours 20 minutes). Only then is the label 'Full-Length' true.
- Delete the entire hardcoded 32-question list from QuestionBank.kt.

6 Out of scope for this phase

Screens, navigation, the Calm Pro theme and tokens, the launcher icon, study-guide content, flashcards, achievements, the leaderboard, scoring logic, and user progress/state handling. Note: any on-device statistics keyed to the old 8-domain taxonomy will reset — acceptable before launch.

7 Acceptance checks (verified in the re-synced code)

- QuestionBank.kt contains no hardcoded questions; a loader reads assets/banks/bank-seed-20261021.json via org.json; build files gained no new dependencies.
- All §5 validation rules are implemented; no fallback question set exists anywhere.
- FsDomain has exactly the 7 entries of §4; the targetExamQuestions field is gone.
- Mock exam: 110 questions, 320-minute limit, proportional domain draw.
- Every screen and feature from the current build is still present and reachable; theme tokens untouched.
- The bank file sits at the §2 install path, byte-identical to the staged payload (SHA-256 in §2).

8 What happens next

After this phase verifies: the GUI simplification (4-tab navigation) and the removal of the 'NCEES' name from all user-facing text, per the UI Simplification Prompt v1.1.0, followed by device QA against the Calm Pro checklist.